Intelligent E-Commerce Customer Analytics Platform

Skills take away From This Project	• Working with Relational Datasets (Multiple CSV Files) • SQL-style Data Merging and Data Engineering • Advanced Exploratory Data Analysis (EDA) • Missing Value Treatment and Outlier Detection • Feature Engineering using Transactional Data • Customer Segmentation (Clustering) • Purchase Prediction (Classification) • Customer Lifetime Value (Regression) • Product Recommendation Systems • Ensemble Learning (Random Forest, XGBoost, LightGBM, CatBoost) • Hyperparameter Tuning • Explainable AI using SHAP • End-to-End Scikit-learn Pipelines • Streamlit Dashboard Development

Domain	**E-Commerce | Customer Analytics | Retail Intelligence | Recommendation Systems**

Problem Statement:

Modern e-commerce businesses generate enormous volumes of transactional data every day. However, simply collecting this data is not enough. Businesses need intelligent systems that can convert customer data into actionable insights for increasing sales, improving customer retention, and delivering personalized shopping experiences.

The objective of this project is to develop an **Intelligent E-Commerce Customer Analytics Platform** that integrates multiple machine learning models into a unified solution. The platform should analyze customer purchasing behavior, segment customers based on buying patterns, predict future purchases, estimate customer lifetime value, and recommend products tailored to individual customers.

The final application should assist marketing teams, business analysts, and decision-makers in making data-driven decisions.

Business Use Cases:

Customer Segmentation

Identify groups of customers with similar purchasing behavior to enable targeted marketing campaigns.

Purchase Prediction

Predict whether a customer is likely to make another purchase within a specified time period.

Customer Lifetime Value Prediction

Estimate future revenue generated by each customer.

Personalized Product Recommendation (Optional)

Recommend products based on customer preferences and purchase history.

Customer Retention Strategy

Identify customers who are becoming inactive and require personalized engagement.

Business Dashboard

Provide interactive visualizations and predictive insights for business users.

Dataset

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Brazilian E-Commerce Public Dataset by Olist

The dataset contains multiple interconnected tables representing a real-world e-commerce ecosystem.

Tables Included

Dataset	Description
customers	Customer information
orders	Order details and timestamps
order_items	Products purchased in each order
order_payments	Payment methods and payment values
order_reviews	Customer ratings and reviews

products	Product metadata and categories
sellers	Seller information
geolocation	Customer and seller locations
product_category_translation	Portuguese to English category mapping

Project Workflow

Phase 1 – Data Integration

Merge all relevant tables to create a customer-centric analytical dataset.

Students should understand

- Primary Keys
- Foreign Keys
- One-to-Many Relationships
- Data Joining Techniques

Phase 2 – Data Cleaning

Perform

- Missing value handling
- Duplicate removal
- Date conversion
- Invalid record handling
- Outlier detection
- Feature consistency checks

Phase 3 – Exploratory Data Analysis

Analyze

- Sales trends
- Monthly revenue
- Category popularity
- Top-selling products
- Customer distribution
- Seller distribution
- Delivery performance
- Payment method analysis
- Review score distribution
- Geographic sales patterns

Visualization examples

- Heatmaps
- Boxplots
- Histograms
- Choropleth Maps (Bonus)
- Time-series plots (Bonus)

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Phase 4 – Feature Engineering

Students should engineer meaningful business features such as:

Customer Features

- Total Orders
- Total Spending
- Average Order Value
- Average Basket Size
- Recency
- Frequency
- Monetary Value (RFM)
- Customer Tenure
- Days Since Last Purchase
- Average Review Rating
- Preferred Payment Method

Product Features

- Product Popularity
- Product Rating
- Category Popularity
- Average Selling Price

Seller Features

- Seller Performance Score
- Average Delivery Time
- Average Review Score

Time-Based Features

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- Purchase Month
- Purchase Weekday
- Purchase Hour
- Seasonal Purchase Indicator
- Festival Sales Indicator (Bonus)

Phase 5 – Customer Segmentation

Objective

Group customers based on purchasing behavior.

Algorithms

- K-Means
- Agglomerative Clustering (Bonus)
- DBSCAN (Bonus)

Evaluation

- Elbow Method
- Silhouette Score
- Visualization
- PCA
- t-SNE (Bonus)

Business Deliverables

- Premium Customers
- Regular Customers
- Bargain Hunters
- New Customers
- Dormant Customers

Phase 6 – Purchase Prediction

Objective

Predict whether a customer will place another order within the next 90 days.

Target Variable

Repeat Purchase

Models any 6 ML models

- Logistic Regression
- Decision Tree
- Random Forest
- Gradient Boosting
- XGBoost
- LightGBM
- CatBoost
- Support Vector Machine
- Naive Bayes

Handling Class Imbalance

- SMOTE
- Class Weighting

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-scor

Phase 7 – Customer Lifetime Value Prediction

Objective

Predict future revenue expected from each customer.

Regression Models any 5 models

- Linear Regression
- Random Forest Regressor
- Gradient Boosting Regressor
- XGBoost Regressor
- CatBoost Regressor
- LightGBM Regressor

Evaluation

- RMSE
- MAE
- MAPE
- R^2 Score

Phase 8 – Product Recommendation System (Optional)

Develop a recommendation engine using one or more of the following approaches:

Content-Based Recommendation

Recommend products based on customer purchase history and product categories.

Collaborative Filtering

Recommend products based on similar customers' purchasing patterns.

Matrix Factorization (Bonus)

Apply techniques such as Singular Value Decomposition (SVD).

Hybrid Recommendation System (Bonus)

Combine content-based and collaborative filtering methods for improved recommendations.

Evaluation Metrics

- Precision@K
- Recall@K
- MAP@K
- Hit Rate

Phase 9 – Explainable AI

Interpret predictions using

- Feature Importance
- SHAP Summary Plot
- SHAP Waterfall Plot
- SHAP Force Plot

Business Questions

- Why is this customer likely to purchase again?

- Which features contribute most to customer value?
- Why was this product recommended?

Phase 10 – Dashboard Development

Develop an interactive Streamlit application with the following modules:

Executive Dashboard

- Total Revenue
- Total Customers
- Orders
- Average Order Value

Customer Analytics

- Customer Segment
- Purchase Probability
- Predicted Lifetime Value
- Customer Profile

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Recommendation Engine (Optional)

- Top 5 Recommended Products

Business Insights

- Sales Trends
- Category Performance
- Seller Performance
- Geographic Sales Distribution

Explainability

- SHAP Visualization
- Feature Importance

Phase 11 – Deployment

Deploy the application using:

- Streamlit

Expected Results

The completed solution should provide:

- Intelligent customer segmentation
- Accurate repeat purchase prediction
- Customer lifetime value estimation
- Personalized product recommendations(Optional)
- Explainable machine learning predictions
- Interactive business dashboard

Evaluation Metrics

Customer Segmentation

- Silhouette Score

Purchase Prediction

- Accuracy
- Precision
- Recall
- F1-score

Customer Lifetime Value

- RMSE

- MAE
- R^2 Score
- MAPE

Recommendation System (Optional)

- Precision@K
- Recall@K
- MAP@K
- Hit Rate

Explainability

- Feature Importance
- SHAP Summary Plot
- SHAP Waterfall Plot

Technical Tags

Machine Learning • Classification • Regression • Clustering • Recommendation Systems • Feature Engineering • Customer Analytics • Business Intelligence • Explainable AI • Ensemble Learning • XGBoost • LightGBM • CatBoost • SHAP • Streamlit • AWS

Project Deliverables

- Integrated and cleaned analytical dataset
- Data integration and preprocessing notebook
- Exploratory Data Analysis notebook
- Feature Engineering notebook
- Customer Segmentation notebook
- Purchase Prediction notebook
- Customer Lifetime Value notebook
- Product Recommendation notebook
- Model comparison report

- Explainability analysis (Feature Importance and SHAP)
- End-to-end Scikit-learn pipelines
- Streamlit dashboard
- Deployed application
- GitHub repository with README
- Final project report
- Presentation demonstrating business insights

Project Guidelines

- Build modular and reusable code using Scikit-learn Pipelines.
- Perform proper train-validation-test splitting and avoid data leakage.
- Document all preprocessing and feature engineering decisions.
- Compare baseline models with advanced ensemble methods.

Use Git for version control and maintain clear project documentation.

Ensure the Streamlit dashboard is intuitive, interactive, and business-focused.

Incorporate explainability techniques to justify model predictions.

Write production-ready, well-documented code following software engineering best practices.

PROJECT DOUBT CLARIFICATION SESSION (PROJECT AND CLASS DOUBTS)

About Session: The Project Doubt Clarification Session is a helpful resource for resolving questions and concerns about projects and class topics. It provides support in understanding project requirements, addressing code issues, and clarifying class concepts. The session

aims to enhance comprehension and provide guidance to overcome challenges effectively.

Note: Book the slot at least before 12:00 Pm on the same day

Timing: Saturday (5:00PM to 7:00PM)

Booking link :https://forms.gle/NtkQ4UV9cBV7Ac3C8

LIVE EVALUATION SESSION (CAPSTONE AND FINAL PROJECT)

About Session: The Live Evaluation Session for Capstone and Final Projects allows participants to showcase their projects and receive

real-time feedback for improvement. It assesses project quality and provides an opportunity for discussion and evaluation.

Note: This form will Open on Saturday and Sunday Only on Every Week

Timing: Monday-Saturday (11:30PM to 12:30PM)

Booking link : https://forms.gle/1m2Gsro41fLtZurRA